FORMULATION

psych scene
RANZCP EXAM PREP

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FORMULATION

Formulation is one of the most integral parts of a psychiatric case. It is this part of the presentation that provides an understanding of the different complexities of patients. Much has been written about psychodynamic formulations, but the main focus of this chapter is on the techniques to generate a formulation using bio-psycho-social models first proposed by Engel. However, with increasing globalisation, migration and, in Australia and New Zealand, indigenous populations, a cultural element is also included and thus the model is a BIO-PSYCHO-SOCIO-CULTURAL model.

TYPES OF FORMULATION

1. Etiological
2. Consequential
3. Speculative

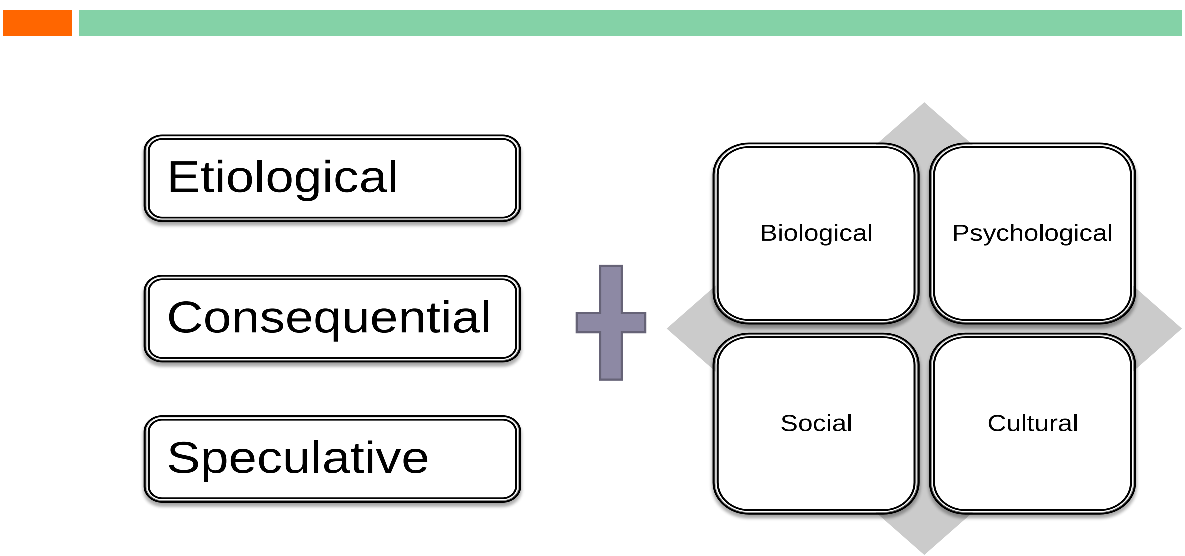
This division is not to suggest that these three types or models are mutually exclusive, but rather they offer a format for thinking about the most useful way of understanding the patient.

COMPONENTS OF FORMULATION

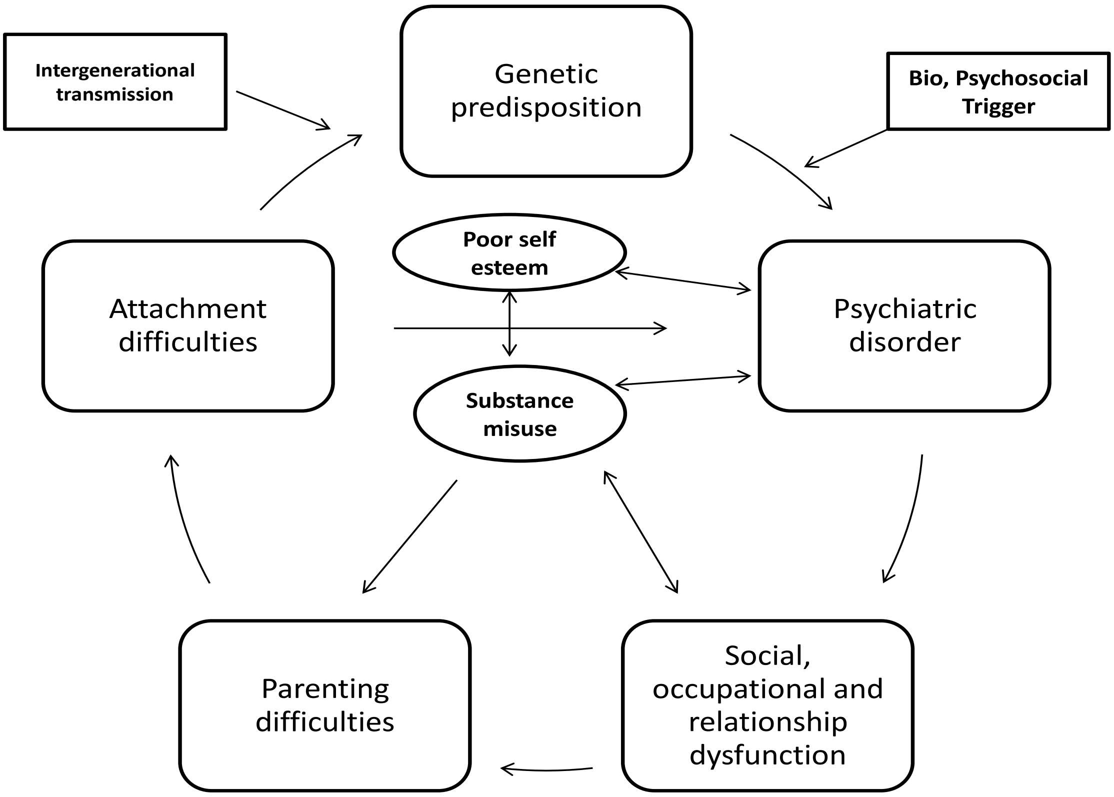
Earlier we looked at the formulation models. Here we look at the individual components of formulation.
The combination of the two gives you a **Formulation matrix**

1. Biological
2. Psychological
3. Social
4. Cultural

Formulation Matrix



MULTIDIRECTIONAL RELATIONSHIP BETWEEN BIOPSYCHOSOCIAL ELEMENTS IN FORMULATION



BIOLOGICAL FORMULATION

There are several aetiologies for psychiatric disorders but the most relevant for formulation and management will be presented here.

1. GENETIC

See table in **Personal History** above, outlining lifetime risks for different psychiatric disorders.

2. NEURODEVELOPMENTAL HYPOTHESIS

This is particularly relevant for schizophrenia and was first proposed by Weinberger. This has been supported by changes in brain structure, pre-morbid cognitive and neuroaffective dysfunction/disabilities, and greater obstetric complications. It is postulated that brain insult in utero or at birth plays a role in expression through interaction with other environmental variables. The neuro-developmental insult also has implications for management as it contributes to treatment resistance. There is accumulating evidence that this hypothesis is relevant for other psychiatric disorders.

3. DRUGS AND ALCOHOL

The above diagram shows that use of drugs and alcohol has a bidirectional relationship with the social, psychological and biological components of the psychiatric disorder. Use of cannabis, alcohol and other drugs is strongly associated with depression and psychosis (Andreasen)

Substance use hypotheses

a) **Self-medication hypothesis** (Khantzian 1985, 1997): Substance used to reduce negative symptoms, affective symptoms and anxiety and depression with mild abuse. Cannabis use is also reported to reduce level of side effects of antipsychotics.

b) **Affect regulation model** (Blanchard 2000): Substance used to modulate affect that may result from individual differences in personality structure and poor coping skills.

c) **Common confounding factor**: There may be an undeveloped confounding factor for cannabis and schizophrenia, or a common genetic basis for cannabis abuse, schizophrenia and underlying neuropsychological vulnerabilities of both disorders.

d) **Motivational Models of Substance Use** (Cox & Kings) These suggest that individuals use substances to achieve desired effects and can encompass affect regulation and social affiliation as motivators for use.

Cannabis

- Cannabis users are more likely to present with alienated psychotic symptoms than non-users.
- Cannabis psychosis may be an independent risk factor for psychosis, especially in subjects with pre-existing vulnerability for such a disorder.
- Cannabis is not a sufficient or necessary cause for schizophrenia, but a component cause which, in combination with other component causes, e.g. genetic predispositions, leads to the development of schizophrenia.
- The biological component of cannabis responsible for psychosis is Tetrahydrocannabinol, which acts on dopaminergic projections of the medial forebrain.

Cannabis use and depression

There are modest associations between early onset regular or problematic cannabis use and depression. The main explanation is biological – neurotransmitter changes. For other causes there is little research evidence – socially modulated.

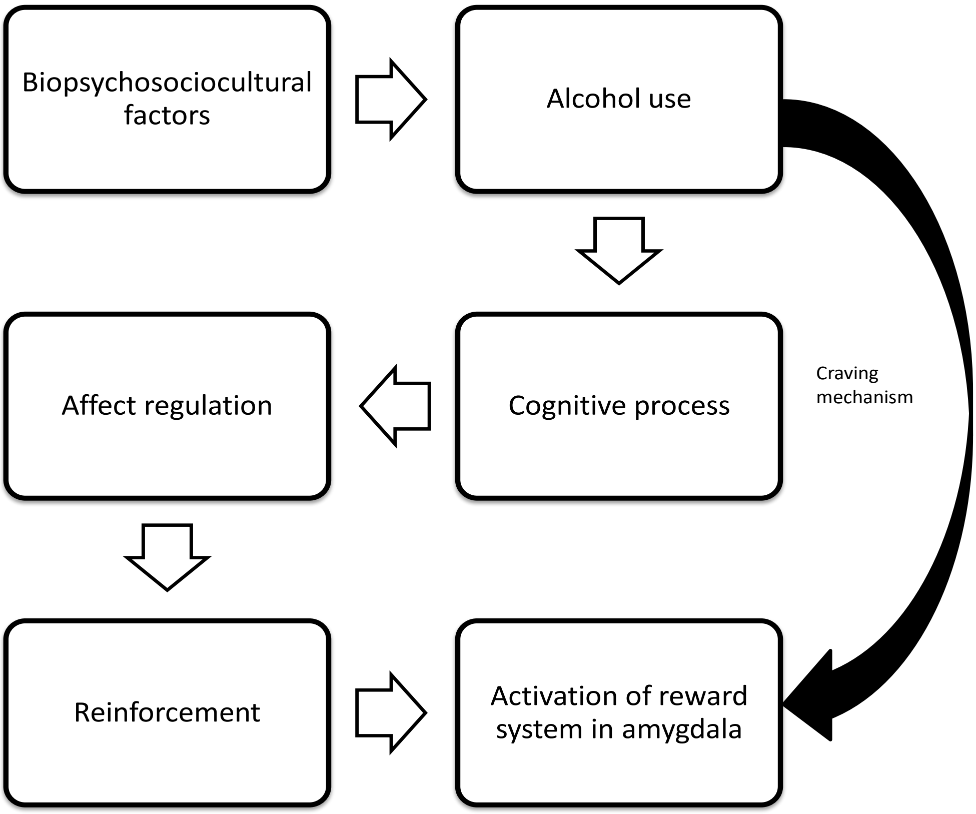
Motives for cannabis use

- **Coping motives**: To reduce and or avoid negative emotional states. To relax, to forget your worries.
- **Social motives**: Facilitate peer group relationships; affiliate with others.
- **Enhancement motives**: To facilitate positive emotions.
  (Newcombe et al. 1988, Johnston & O'Malley, Fowler 1998)

Simons (2000) showed that motives of enhancement, coping and conformity were common for both drug and alcohol use and that further effects were affect regulators.

ALCOHOL

AFFECT REGULATION HYPOTHESIS IN ALCOHOL



4. EPILEPSY

Composite hypothesis (Sachdev, AJP, 1998)

Neurodevelopmental dysgenesis particularly in limbic region (trauma, hypoxia)

↓

Electrical discharges in limbic cortex leading to frank seizures or micro seizures

↓

Kindling mechanisms leading to ongoing seizures preventing axonal and synaptic regeneration and reorganisation

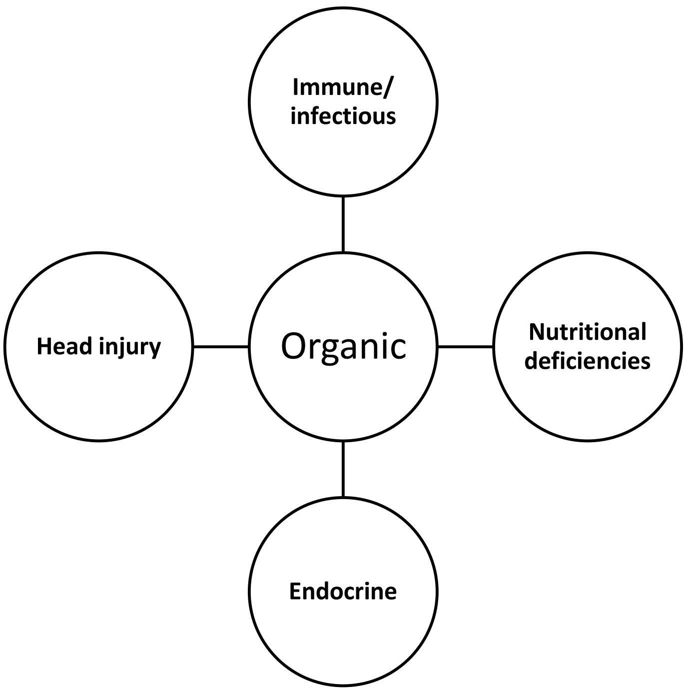
↓

Underlying physiological and anatomical changes leading to psychosis

↓

Further complicated by Long-Term Drug therapy (neurotoxicity) and psychosocial factors related to epilepsy

ORGANIC DISORDERS



PSYCHOLOGICAL FORMULATION

1. ATTACHMENT MODEL

The attachment theory was first proposed by Bowlby. It is primarily a process of seeking proximity to an identified attachment figure in stressful situations, for the purpose of survival. Mary Ainsworth identified different attachment styles, e.g. secure, insecure and disorganized attachment.

The most relevant points concerning the attachment model and the genesis of personality disorders or maladaptive personality traits are described here:

George & Solomon (1996) and Schore postulated that a distressed infant responds to threats in their environment by experiencing a high degree of arousal, mediated by the sympathetic nervous system which in turn results in high blood pressure, heart rate and respiratory rate, which are unpleasant feelings. The mother's role in soothing the infant is to recruit the baby's peripheral nervous system to restore homeostasis. A good quality affect regulatory system, based on secure attachment, leads to optimal maturation of the right hemisphere at a critical period during the first 2–3 years of life. Any experience that disturbs this attachment will lead to impaired development of neural pathways that subserve emotional behaviours, leading to impaired emotional regulation that may persist through life. These experiences include childhood sexual abuse, physical trauma and loss (Schore 2002).

Individuals with personality disorder have significant difficulty in maintaining relationships for these reasons, and this is often evident in the doctor/patient relationship, in which the doctor is the care giver.

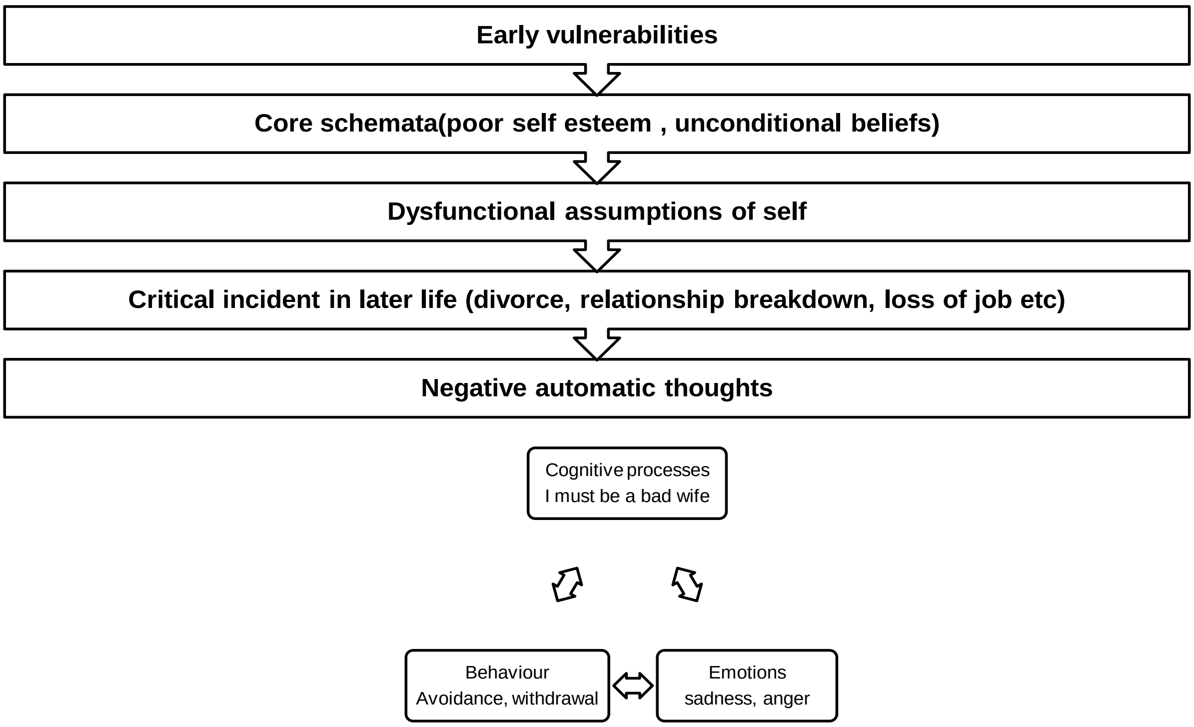
Thus, attachment difficulty forms a vulnerability factor that predisposes the individual to poor self-esteem and the development of dysfunctional assumptions of self-cognition. It inhibits children from developing the internal and interpersonal coping skills needed to buffer against family, social and community stressors that can cause or exacerbate depression.

2. COGNITIVE BEHAVIOURAL MODELS

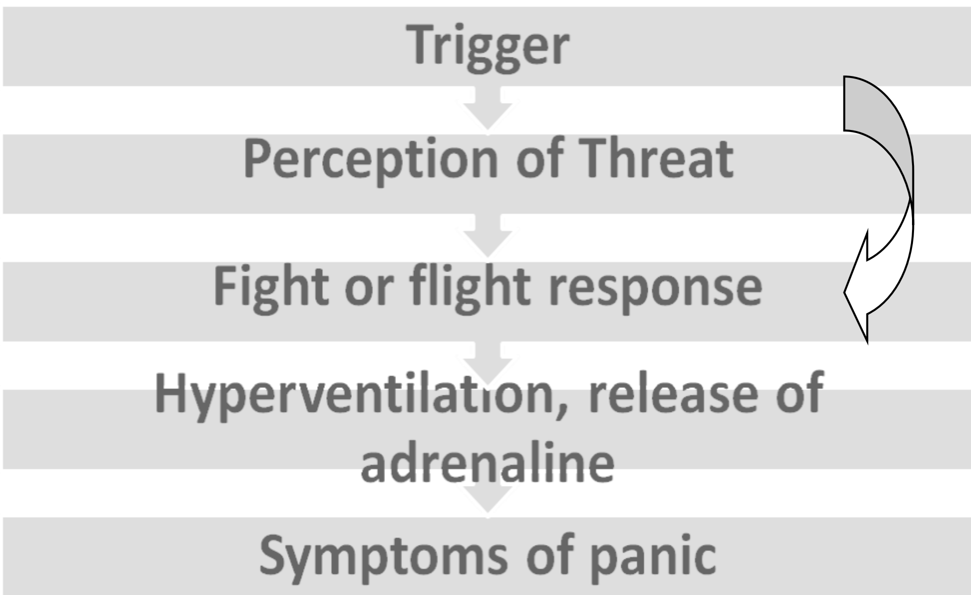
Cognitive behavioural models of formulation are especially useful as they look at the here and now and provide a simple way of understanding that links in well with the management plan. Cognitive models are useful in depression, anxiety, bipolar disorder and bulimia nervosa.

COGNITIVE MODEL IN DEPRESSION

Cognitive Model: Depression



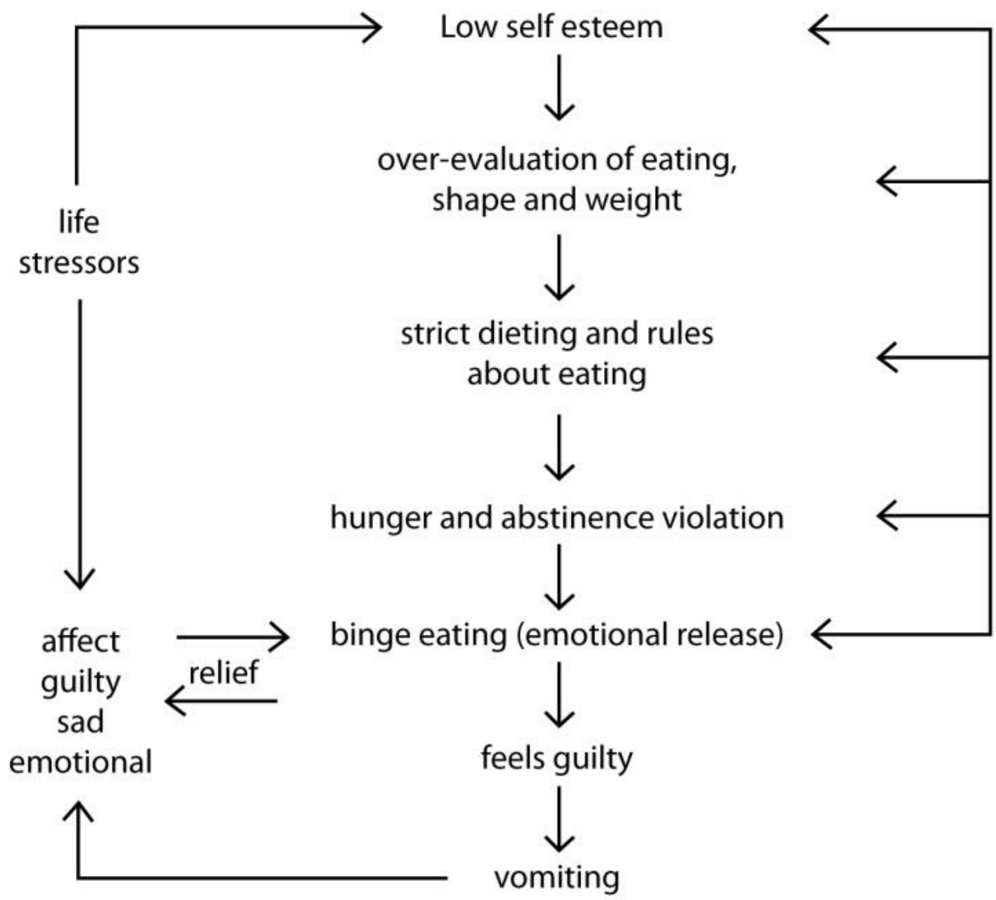
ANXIETY MODEL



BIPOLAR DISORDER

In Bipolar Disorder, the cognitive behavioural model is based on a diathesis-stress model. It proposes that there is an inherent psychological vulnerability (genetic predisposition) as a function of instability of circadian rhythms and of impaired regulation of the motivational system controlling approach and reward. Certain events in early life may result in the development of low self-esteem and the desire to strive for affection. This may lead to the development of unrelenting standards for success that, if not achieved, predispose the individual to depression. On the other hand, the striving for success and affection may, from a biological and psychological point of view, disrupt social and circadian rhythms, contributing to a manic syndrome.

COGNITIVE MODEL FOR BULIMIA NERVOSA



3. ERIKSON'S PSYCHOSOCIAL DEVELOPMENT

Erikson's epigenetic model using the 'eight ages of man' covers the whole life cycle and is particularly useful in formulations. Although this can be applied to individuals at all ages, it is most relevant and useful in the elderly. Other examples would include difficulty with separation, and individuals in enmeshed family relationships.

4. PSYCHODYNAMIC FORMULATIONS

Only the relevant ones are mentioned here. Beware, however of making psychodynamic interpretations in the acutely psychotic patient.

- **Splitting** – Borderline Personality Disorder. Splitting individuals into the "good mother" and "bad mother" can result in difficulties in engagement.
- **Projective identification** – Project feelings of helplessness, anger and hostility leading to difficulties in management.
- **Mania** – Defence against depression.
- **Conflicts**_– Achieving independence vs. dependence on family. Common in Schizophrenia.
    - Identity conflicts – Borderline personality (also due to impaired object relationships)
      Desire for care vs. autarchy (Borderline Personality Disorder)

5. SOCIAL SKILLS MODEL

This proposes that social dysfunction arises from skills deficit and is linked with stress vulnerability. Coping skills model, i.e., social skills are a protective factor limiting the negative impact of stress. The failure to acquire social skills at crucial stage of early development, e.g., adolescence, can lead to disturbed peer group and intimate relationships which in turn can lead to psychological distress and social isolation.

The other social aspects to be considered are housing, finances, relationships and employment, all of which can precipitate and maintain a psychiatric disorder. In view of this, clinicians should incorporate these aspects in management by involving members of multidisciplinary teams.

SAMPLE FORMULATION- CASE 1

From a **biological point of view**, X has a genetic predisposition to developing a mood disorder as evidenced by a family history of affective illness. Additional biological contributing factors to a mood disorder include her cannabis misuse and Sumatriptan. In addition, she also reported a significant head injury where she lost consciousness which may have resulted in brain injury resulting in neuropsychiatric symptoms including worsening of her affect dysregulation.

From a **psychological point of view**, an important thing that arises is the presence of parental criticality, excessive focus on weight and appearance by her parents when she was a child which would have led to dysfunctional assumptions of self which have been reinforced and reactivated through her late life relationship difficulties. Her vocational instability over the years with unstable relationships has possibly reinforced her negative automatic thoughts thus contributing to her depression.

Her mood and emotional dysregulation can also be conceptualised within a **dialectical behavioural model** where her underlying biological vulnerability to emotional dysregulation along with her emotionally invalidating environment, has resulted in a pervasive emotional dysregulation for her at this point in time.

Additionally, there are several **interpersonal aspects** that can also be considered which include grief (loss of her grandfather at an early age and loss of her partner 10 years ago), role transitions (loss of job and interpersonal sensitivity, inability to maintain significant relationships). It is possible that her dysfunctional attitude to pain and its consequences may stem from possible role models in childhood; however, this was not positively identified. It is possible that this abnormal illness behaviour i.e. sick role may be leading to secondary gain. She has in addition seen a number of doctors and I wonder if there are engagement issues that would impact on a therapeutic relationship.

Thus, her illness can be conceptualised within the above psychological and biological factors. Her depression and mood instability are being maintained by her prominent cognitive dysfunctional assumptions, relationship and vocational instability.

SAMPLE FORMULATION - CASE 2

From a **biological point of view**, Y has a number of biological predisposing factors that contribute to her illness. Particularly relevant is the presence of a strong genetic predisposition in the family towards both Bipolar affective disorder and alcohol dependence. The presence of polycystic ovarian disease which was diagnosed in 1995 is also a significant contributory factor to the affective illness along the lines of the hormonal hypothesis.

Also, of note is the presence of obstetric complications which is a predisposing factor for a psychotic illness from a neurodevelopmental hypothesis. Additional biological contributing factors include possible head injury at the age of 8, diabetes which is a vascular contributory factor towards depression (vascular hypothesis of depression). She is currently on a combination of antipsychotics and her chewing movements may be considered as tardive dyskinesia as antipsychotics are known to increase the risk of tardive dyskinesia in individuals with Bipolar affective disorder. However, along the same lines, I am concerned about the presence of parkinsonian symptoms and wonder if the dyskinetic movements indicate an additional organic contributor to the illness.

From a **psychological point of view**, an important finding is the presence of early childhood maladjustment due to alcohol dependence in her father and domestic violence towards her brother in the household. This has most likely led to attachment difficulties, contributing to

low self-esteem, which has manifested in dysfunctional assumptions about herself, thus setting up a vulnerability for future depressive episodes. Her mania can be conceptualised as a defence against her depression.
Also relevant is the onset of illness at an early age with particular difficulties in high school related to education, prominent anxiety and I wonder whether this was a prodrome which has impacted on her early psychosocial development.

Thus, her illness can be understood in the context of the above psychological and biological factors. Her illness is being maintained currently by significant psychosocial stressors, anxiety about going to work, ongoing organic contributors, side effects of the medication and relationship instability. Fortunately, there are strengths such as engagement and compliance.